



Mutual Fund Performance Analytics — 2024

Exploratory Data Analysis & Operational Exception Framework

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Domain: Investment Performance Measurement & Analytics

Tools: Python (Pandas, NumPy), Visualization (Matplotlib, Seaborn)

Project Overview

This project analyses 2024 mutual fund return data to:

- Standardize and ingest complex monthly performance records.
- Apply dynamic operational boundaries to identify performance exceptions (HIGH / LOW / NORMAL).
- Map category-wise performance distributions and uncover macro market trends.

Data Loaded Successfully: 1646 funds across 15 columns.

	Fund	Fund_Manager	Category	Mar-24	Apr-24	May-24	Jun-24	J
0	360 ONE Balanced Hybrid Fund Reg(G)	Mayur Patel	HY-EQ	0.61	3.19	2.60	4.30	2
1	360 ONE Dynamic Bond Fund Reg(G)	Milan Mody	DT-DYN	0.65	-0.05	0.97	0.66	1
2	360 ONE ELSS Tax Saver Nifty 50 Index Fund Reg(G)	Parijat Garg	EQ-ELSS	1.51	1.20	-0.02	6.66	3
3	360 ONE Flexicap Fund Reg(G)	Mayur Patel	EQ-FLEX	0.61	6.66	4.16	8.27	3
4	360 ONE Focused Equity Fund Reg(G)	Mayur Patel	EQ-MLC	0.76	5.48	1.66	6.75	4

=== Performance Range Distribution ===

Peak Average Return: 6.80%
Systemic Portfolio Mean: 1.04%
Median Performance: 0.90%

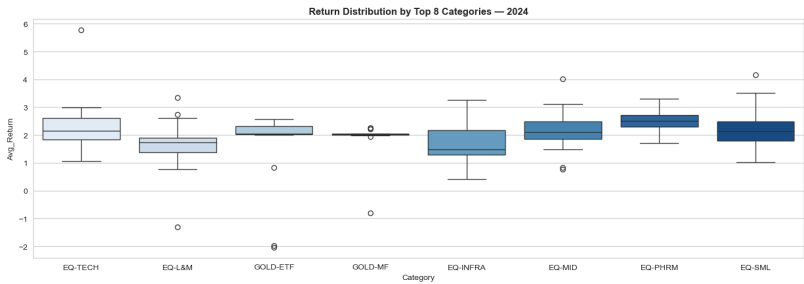
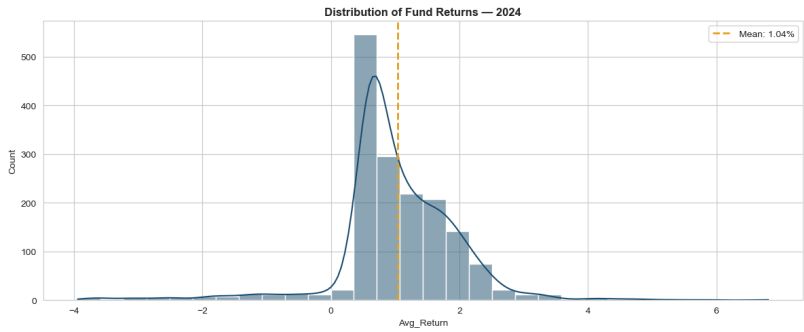
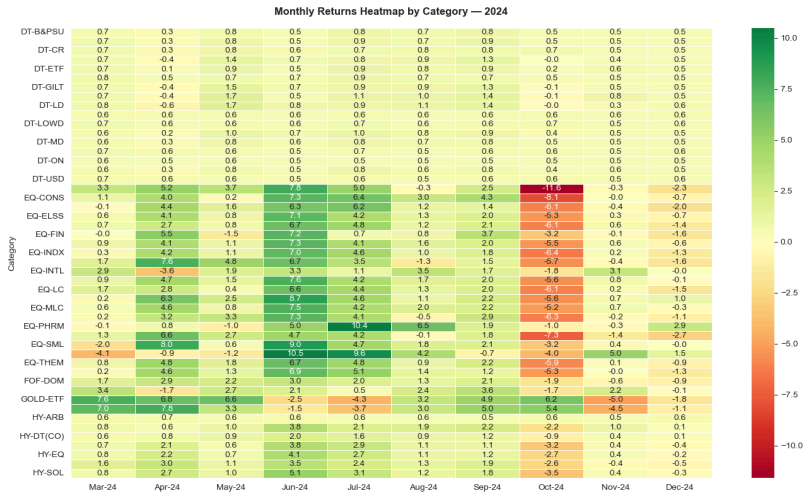
=== Monthly Market Performance Trend ===

Best Calendar Month: Jun-24 (4.14%)
Worst Calendar Month: Oct-24 (-2.96%)

=== Exception Verification Summary ===

Exception_Flag
NORMAL 1547
LOW EXCEPTION 85
HIGH EXCEPTION 14
Name: count, dtype: int64

✔ Audited data successfully exported to:
C:\Users\bidar\OneDrive\Desktop\Project2-EDA\performance_summary.csv



5. Key Findings

- **Total Funds Analysed:** 1,646 mutual funds across 49 unique asset categories.
- **Top Performing Categories (Mean Return):** EQ-PHRM (Pharma) and EQ-TECH (Technology).
- **Market Exception Distribution:**
 - **High Performance Exceptions:** 14 funds exceeded the dynamic alpha threshold ($>3.5\%$ monthly average).
 - **Low Performance Exceptions:** 85 funds fell below the downside risk floor ($<0\%$ monthly average).
 - **Normal Performers:** 1,547 funds maintained market-conforming steady returns.



Institutional Performance Insight

Performance exceptions were identified using dynamic threshold boundaries tailored specifically to the dataset's distribution maximum (6.80%) and median (0.90%). This mirrors real-world exception investigation frameworks used by institutional investment operations to isolate elite alpha generators and flags operational exceptions for further auditing.

Starting report compilation...